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Inventor(s)

Shields; Matthew et al.

SYSTEM AND METHOD FOR FIBERGLASS REINFORCED PANEL

Abstract

A self-bonding panel comprising a fiberglass reinforced panel having a front side and a back side. At least one adhesive layer is applied to the backside of the fiberglass reinforced panel. At least one release liner layer is applied to a surface of the at least one adhesive layer opposite the fiberglass reinforced panel.

Inventors: Shields; Matthew (Chicago, IL), Butman; Jacob (Highland Park, IL), Kirk; Sharon (New Lenox, IL), Paramo; Alan (Joliet, IL), Yuede; Nicole (Oakbrook Terrace, IL)

Applicant: Crane Composites, Inc. (Channahon, IL)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION AND PRIORITY CLAIM [0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/562,590 filed on Mar. 7, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The following disclosure relates to fiberglass reinforced panels, and more particularly, to fiberglass reinforced panels having a self-bonding surface.

BACKGROUND

[0003] Fiberglass reinforced panels (FRPs) are widely used within the construction industry. Normal application of FRPs require the application of a wet trowel adhesive to a backside of the FRP prior to its application to a substrate. The substrate may comprise in some implementations a gypsum wallboard or other solid substrates that also require the application of a wet trowel adhesive. After application of the FRP to the substrate it may be moved around to provide a final location position. The wet trowel adhesive will then dry, securing the FRP in place on the substrate. The application of FRPs in this manner requires a great deal of labor and time in the creation of the wet trowel adhesive used to secure the FRPs to a substrate, the application of the wet trowel adhesive to a backside of the FRP and the positioning of the FRP on a substrate. The current application process of FRPs creates a labor and time intensive process at the construction site. Thus, a structure and method for improving the speed and efficiency for the application of FRPs to substrates would provide a great benefit with the associated savings in both manpower and time and provide uniformity and consistency in adhesive applying the FRP.

SUMMARY

[0004] The present disclosure illustrates embodiments of various self-bonding FRPs.

[0005] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims. A self-bonding panel comprises a fiberglass reinforced panel having a front side and a back side. At least one adhesive layer is applied to the backside of the fiberglass reinforced panel. At least one release liner layer is applied to a surface of the at least one adhesive layer opposite the fiberglass reinforced panel.

[0006] Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The term “couple” and its derivatives refer to any direct or indirect interaction between two or more elements, whether or not those elements are in physical contact with one another. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

[0007] Definitions for other certain words and phrases are provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings in which like reference numerals represent like parts:

[0009] FIG. 1 illustrates an example of an application of multiple FRPs to a structure in accordance with various embodiments of the present disclosure;

[0010] FIG. 2 illustrates a view of proposed embodiments of the multiple layers of an FRP in accordance with various embodiments of the present disclosure;

[0011] FIG. 3 illustrates the application of an FRP to a substrate after having its release liner layer removed in accordance with various embodiments of the present disclosure;

[0012] FIG. 4A illustrates a first embodiment of an FRP, in accordance with various embodiments of the present disclosure;

[0013] FIG. 4B illustrates an alternative embodiment of the first embodiment of the FRP;

[0014] FIG. 5 illustrates a second embodiment of an FRP, in accordance with various embodiments of the present disclosure;

[0015] FIG. 6 illustrates a third embodiment of an FRP, in accordance with various embodiments of the present disclosure;

[0016] FIG. 7 illustrates a fourth embodiment of an FRP, in accordance with various embodiments of the present disclosure;

[0017] FIG. 8 illustrates an FRP connected to a substrate in accordance with various embodiments of the present disclosure;

[0018] FIG. 9 illustrates a division bar trim accessory for installing FRPs in accordance with various embodiments of the present disclosure;

[0019] FIG. 10 illustrates an endcap trim accessory for installing FRPs in accordance with various embodiments of the present disclosure;

[0020] FIG. 11 illustrates an inside corner trim accessory for installing FRPs in accordance with various embodiments of the present disclosure;

[0021] FIG. 12A-12C illustrate a top view of the adhesive and release liner layers laminated to the FRP; and

[0022] FIG. 13 illustrates a flow diagram of the installation of the self-bonding FRP to a substrate.

DETAILED DESCRIPTION

[0023] FIGS. 1 through 13, discussed below, and the various embodiments used to describe the principles of the present disclosure are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. The figures are not necessarily drawn to scale, and in some instances the drawings have been exaggerated and/or simplified in places for illustrative purposes only. One of ordinary skill in the art will appreciate the many possible applications and variations based on the following examples of possible embodiments.

[0024] FIG. 1 illustrates the manner in which individual fiberglass reinforced panels (FRPs) **102** are connected to a structure **104**. The structure **104** may comprise a substrate on which an FRP **102** may be mounted. This may include walls, ceilings, floors, ducts, or any other flat substrate to which the FRP **102** may be mounted. The FRP **102** may be used in construction of buildings, homes, recreational vehicles, aircraft, or any other application that would need the application of an FRP **102** to any substrate. Each of the FRPs **102** are secured to the structure **104** in the manner which will be more fully described hereinbelow. FRP **102A** is shown prior to its securing to the structure **104** and is placed in a position adjacent to other FRPs **102** on the structure **104** so that it may be secured in place. While the present disclosure describes the FRP **102** as comprising a single panel, the implementation of the FRP **102** may also comprise a roll or coil of FRP material that

may be rolled out along a substrate and then cut to a desired length before installation as described herein below.

[0025] FIG. 2 illustrates a self-bonding FRP 202. The self-bonding FRP 202 comprises three separate layers. The process for creating the self-bonding FRP 202 laminates an adhesive layer 206 to the back side of an FRP panel 204. These three separate layers comprise the FRP 204, the adhesive layer 206, and the release liner layer 208. Added to the FRP panel 204 are the adhesive layer 206 and the release liner layer 208. The adhesive layer 206 enables the FRP 204 to be secured to a substrate. The adhesive layer 206 is pressure sensitive such that FRP 204 may be placed in a desired initial position and finally secured to a structure by applying pressure to a front surface of the FRP 204. The release liner layer 208 comprises a removable film, poly coated kraft paper, direct coated paper etc. which protects adhesive layer 206 until the FRP 204 is to be secured to a structure. While the preferred embodiment uses a pressure sensitive adhesive, other types of adhesives may be used if the adhesive provides the adhesive layer 206 the ability to bond to both of the surfaces that the FRP panel 204 is being bonded to and to the FRP 204. Additionally, rather than comprising a pressure sensitive adhesive, the adhesive layer 206 may comprise a moisture cure adhesive or other type of adhesive that is activated by any means enabling the FRP 204 to be mounted to a particular substrate.

[0026] FIG. 3 illustrates the self-bonding FRP 202 that is being secured to a particular substrate 302. In the illustration of FIG. 3, the release liner layer 208 has been removed from the backside 304 of the adhesive layer 206. The release liner layer 208 may then be disposed of. The backside 304 of the adhesive layer 206 is placed upon the substrate 302. As mentioned previously, the substrate 302 may comprise any substrate that may need coverage with the self-bonding FRP 202. Once the desired position for the FRP 204 is achieved, a pressure may be applied to the front surface 306 of the self-bonding FRP 202 to secure the self-bonding FRP in its final desired position upon the substrate 302. Activation may occur in other ways corresponding to the adhesive used (i.e., pressure for PSA, water for moisture cured, etc.). If repositioning is desired, the self-bonding FRP 202 can be removed and repositioned on the substrate 302. A self-bonding FRP 202 of this type will reduce labor and the time required to install as compared to current methods for installing a FRPs using a wet trowel adhesive.

[0027] FIG. 4A illustrates a first embodiment of a self-bonding FRP 402 in accordance with various embodiment of the present disclosure. The portion that is applied to the substrate includes the FRP 404 having a number of adhesive layers 406 applied to a backside thereof. The adhesive layers 406 consist of an adhesive A layer 408 applied directly to the backside of the FRP 404, a carrier layer 410 located on top of the adhesive A layer 408, a second adhesive B layer 412 consisting of adhesive located on top of the carrier layer 410. When the adhesive B layer 412 contacts a substrate of the structure, the adhesive B layer 412 maintains a fixed position of the FRP 404 to the structure. Adhesive A layer 408 comprises a permanently bondable solvent-based-adhesive for application to the backside of the FRP 404.

[0028] In one embodiment, the carrier layer 410 comprises a foam carrier that can range from 0.1 mm-200 mm in thickness and provides a number of valuable qualities to adhesive layers 406. Due to the qualities such as resilience, excellent buoyancy, good thermal insulation, excellent strength and shock absorption, low water absorption, impervious to mildew, mold, rot, and bacteria, excellent chemical resistance, non-toxic, and contains no CFCs, HCFCs, or hydrocarbon blowing agents. Carrier layer 410 has smooth skin on both sides and incredibly fine closed cells. The physical attributes carrier layer 410 include lightness, flexibility, softness to the touch, strength, toughness and resistance to moistures, chemicals and extreme temperatures. Carrier layer 410 typically provides improved dimensional consistency and stability over a wide range of fabrication techniques and end use conditions. Carrier layer 410 additionally provides superior thermal stability and insulating properties. The high buoyancy, low moisture permeability and ability to effectively insulate against sound and vibration make polyolefin foams useful in the building and

construction sector. Additionally, the carrier layer **410** is UV and fungus resistant. The carrier layer **410** can withstand service temperatures as low as -80°C . and as high as 100°C . Other types of materials may be used for the carrier layer **410** including Paper, Cloth, Felt, Foam, Metal Foil, Plastic Film/Polymer, PET/Polyester, Polyimide, PVC, Rubber, Silicone, Acrylic, Glass/Fiberglass, Filament and Fluoropolymer/PTFE/PVDF. The adhesive layers **406** that will be applied directly to the substrate to secure the self-bonding FRP **402** comprise a high performance repositionable solvent based adhesive. The adhesive layers **406** allows for the removal and repositioning of the self-bonding FRP **402** against multiple substrates to provide a permanent bond.

[0029] When the self-bonding FRP **402** has yet to be applied to a substrate, the adhesive B layer **412** is protected by a release liner layers **418**. The release liner layers **418** comprise a double coated tape applied to the back side of the self-bonding FRP **402** on the adhesive layers **406** providing a poly-coated paper, film, or paper siliconized release liner. The release liner layers **418** further comprise a tight release coating 1 layer **420** and an easy release coating 2 layer **422** that are separated by a natural or bleached polyethylene kraft, film, or paper differential release liner **424**. When used, the release liner layers **418** are removed from the adhesive B layer **412** and expose the adhesive B layer **412** such that it may be used to secure the self-bonding FRP **402** to a structure. The release liner coatings may comprise any low surface energy coating using in industry, such as, silicone, fluoropolymers, carbamates, polyacrylates, polyolefins, fluorocarbons, chromium stearate complexes, silicone containing copolymers, polymers with long alkyl or fluoroalkyl side chains, etc. While the present description is envisioned with respect to connecting the FRP **404** to a substrate, the FRP **404** may be connected to any appropriate structure.

[0030] Referring now to FIG. **4B**, there is illustrated an alternative embodiment of the self-bonding FRP **402** of FIG. **4A**. In this embodiment, the structure of the self-bonding FRP **402** is the same with the exception that the release coating 2 layer **422** is not present. In this embodiment, the release liner **424** remains with the self-bonding FRP **402** until removed to enable application of the self-bonding FRP to a substrate.

[0031] FIG. **5** illustrates a second embodiment of a self-bonding FRP **502** in accordance with various embodiments of the present disclosure. The portion that is applied to the substrate includes the FRP **504** having a number of adhesive layers **506** applied to a backside thereof. The adhesive layers **506** consist of an adhesive A layer **508** applied directly to the backside of the FRP **504**, a carrier layer **510** located on top of the adhesive A layer **508**, a second adhesive B layer **512** consisting of adhesive, with a performance layer **514** located on top of the second adhesive B layer **512** which has a third adhesive C layer **516** located on top of the performance layer **514**. The performance layer **514** provides cohesive strength between the adhesive layers **516** and **512** and may comprise tissue or other supporting material. When the adhesive C layer **516** contacts a substrate of the structure, the adhesive C layer **516** maintains a fixed position of the FRP **504** to the structure. Adhesive A layer **508** comprises a permanently bondable solvent-based adhesive for application to the backside of the FRP **504**.

[0032] In one embodiment, the carrier layer **510** comprises a foam carrier that can range from 0.1 mm-200 mm in thickness and provides a number of valuable qualities to adhesive layers **506**. The carrier layer **510** has characteristics similar to those discussed above with respect to FIG. **4**.

[0033] The adhesive layers **506** that will be applied directly to the structure to secure the self-bonding FRP **502** is a high performance repositionable solvent based adhesive. The adhesive layer **506** allows for the removal and repositioning of the self-bonding FRP **502** against multiple substrates to provide a permanent bond.

[0034] When the self-bonding panel **502** has yet to be applied to a substrate, the adhesive layer C layer **516** is protected by a release liner layers **518**. The release liner layers **518** comprise a double coated tape applied to the back side of the self-bonding FRP **502** on the adhesive layers **506** providing a poly-coated paper, film, or paper siliconized release liner. The release liner layers **518** further comprise a tight release coating 1 layer **520** and an easy release coating 2 layer **522** that are

separated by a natural or bleached Polyethylene kraft, film, or paper differential release liner **524**. When used, the release liner layers **518** are removed from the adhesive layers **516** or **512** (the outermost adhesive layer) and expose the adhesive C layer **516** such that it may be used to secure the self-bonding FRP **502** to a structure. While the present description is envisioned with respect to connecting the FRP **504** to a substrate, the FRP **504** may be connected to any appropriate structure. [0035] FIG. **6** illustrates a third embodiment of a self-bonding FRP **602** in accordance with various embodiments of the present disclosure. This embodiment includes substantially the same layers as that previously discussed with respect to FIG. **4A**. Additionally, a texture liner layer **626** is included between release liner layer **624** and the release coating 1 layer **620**. Texture layer **626** comprises a clear printed texture or structured embossment of poly layer within the release liner to mechanically enhance repositionability of adhesive to substrate by imparting peaks and valleys into adhesive B layer **612** for less intimate contact prior to applying pressure. The texture layer **626** may also be either embossed or printed onto the release liner **624**. The texture layer **626** may also comprise an embossed film that comprises a coated layer having particles in the layer (a matte surface with texture).

[0036] FIG. **7** illustrates a fourth embodiment of a self-bonding FRP **702** in accordance with various embodiments of the present disclosure. The portion that is applied to the substrate includes the FRP **704** having a number of adhesive layers **706** applied to a backside thereof. The adhesive layers **706** consist of a first adhesive layer **708** applied to the backside of the FRP **704**, a carrier layer **710** located on top of the first adhesive layer **708**, a second adhesive layer **712** consisting of adhesive located on top of the carrier layer **710**, and an performance layer **714** located on top of the second adhesive layer **712**, which has a third adhesive C layer **716** located on top of the performance layer **714**, if present. When the adhesive C layer **716** contacts a substrate of a structure, the adhesive C layer **716** maintains a connected position of the FRP **704** to the substrate. Adhesive layer **708** comprises a permanently bondable solvent-based adhesive for application to the backside of the FRP **704**. The adhesive layer to be applied to the structure or other substrates is high performance repositionable solvent adhesive.

[0037] In one embodiment, carrier layer **710** comprises a foam carrier that can range from 0.1 mm-200 mm in thickness and provides a number of valuable qualities to the adhesive layers **706**. The carrier layer **510** has characteristics similar to those discussed above with respect to FIG. **4**.

[0038] When the adhesive panel has yet to be applied to a structure, the adhesive C layer **716** is protected by a release liner layers **718**. The release liner layers **718** comprise a double coated tape applied to the back side of the FRP **704** on the adhesive layers **706** providing a poly-coated kraft, film, or paper release liner. The liner layers **718** comprise a tight release coating 1 layer **720** and an easy release coating 2 layer **722** that are separated by a poly-coated Kraft, film, or paper release liner **724**. Texture layer **726** comprises a clear printed texture or structured embossment of poly layer within the release liner to mechanically enhance repositionability of adhesive to substrate by imparting peaks and valleys into adhesive C layer **716** for less intimate contact prior to applying pressure. The texture layer **726** may also be either embossed or printed onto the release liner **724**. The texture layer **726** may also comprise an embossed film that comprises a coated layer having particles in the layer (a matte surface with texture).

[0039] When removed, the release liner layers **718** are removed from the adhesive layers **712** or **716** and expose the adhesive C layer **716** such that it may be used to secure the FRP **704** to a structure. While the present description is made with respect to connecting the FRP **704** to a wall substrate, the FRP **704** may be connected to any appropriate structure.

[0040] Referring now to FIG. **8**, there is illustrated the self-bonding FRP **402** of FIG. **4** and one manner that the panel is mounted to a substrate **802**. Once the release liner layers are removed from the adhesive layers **406** mounted to the back surface of the FRP **404**, an adhesive B layer **412** is exposed. The adhesive B layer **412** may be secured to a substrate **802** to secure the FRP **404** to the substrate **802**. If the self-bonding FRP **402** is merely placed on the substrate **802** without applying a

surface pressure to the FRP **404** in the direction generally shown by the arrows **806**, the self-bonding FRP **402** can be easily repositioned by pulling the panel off of the substrate **802** and repositioning the panel to a desired position. Once a final position of the self-bonding FRP **402** has been achieved, pressure may be placed on the front surface of the FRP **404** to secure the adhesive layer **B 412** to the substrate **802**.

[0041] FIG. **9** illustrates a division bar trim accessory **902** for speeding the installation process of a self-bonding FRP **402**. The division bar trim accessory **902** enables the interconnection of two adjacent self-bonding FRPs **402** on a substrate. The division bar trim accessory **902** includes a first portion **904** that connects to a substrate and a second portion **906** that connects with and secures edges of a pair of adjacent self-bonding FRPs **402** on a substrate. The first portion **904** includes a backplate **908** that connects to a substrate. At least a portion of a back side of the backplate **908** includes an adhesive **910** that is used for securing the backplate **908** of the first portion **904** to the substrate. In one embodiment, the adhesive **910** only covers half of the back side of the backplate **908** (other areas may be covered). The remaining half of the backside of the backplate **908** includes no adhesive **910** to allow the non-adhesive portion of the backplate to tuck behind an existing FRP **404** on a substrate. While covering half of the back side of the backplate **908** with adhesive **910** has been described, it will be understood that lesser and greater areas on the back side of the backplate **908** may be covered with the adhesive **910**.

[0042] The front side of the backplate **908** includes edge markers **912** marking the edge placement of a self-bonding FRP **402**. The front side of the backplate **908** of the first portion **904** has a female connector **914** extending perpendicularly therefrom. The female connector **914** includes a pair of latching members **916** extending perpendicular from the backplate **908**. Each of the pair of latching members **916** define a ledge **918** for securing a male connector **920**. The latching members **916** may flexibly move laterally to allow disengagement of the male connector **920**.

[0043] The second portion **906** of the division bar trim accessory **902** comprises a flexible plate **922** and the male connector **920** extending therefrom. The flexible plate **922** has a slightly curved shape where the edges extend downward toward the first portion **904** for holding edges of adjacent self-bonding FRPs **402** against the front surface of the backplate **908**. The flexible nature of the flexible plate **922** enables the division bar trim accessory **902** to accommodate multiple self-bonding FRP **402** thicknesses. The flexible nature further enables the flexible plate **922** to provide a holding force against the edges of the adjacent self-bonding FRPs **402**.

[0044] The male connector **920** includes a horizontal member **924** extending perpendicularly from the surface of the flexible plate **922**. At the end of the horizontal member **924**, opposite from the connection point with the flexible plate **922**, an arrow shaped connector includes a first wing **926** and a second wing **928** enabling the male connector **920** to latch into the female connector **914**. When snapped into place, the first and second wings **926**, **928** click into place on the ledges **918** of the female connector **914** to secure the flexible plate **922** to the backplate **908**. This enables the division bar trim accessory **902** to secure edges of self-bonding FRPs **402** to a substrate. The division bar trim accessory **902** speeds up install times and enhances ease of use of the self-bonding FRP **402**.

[0045] FIG. **10** illustrates an endcap trim accessory **1002** for speeding the installation process of a self-bonding FRPs **402**. The endcap trim accessory **1002** enables the securing of an edge of self-bonding FRPs **402** when not adjacent to an additional self-bonding FRP and providing an endcap for protecting the edge. The endcap trim accessory **1002** includes a first portion **1004** that connects to a substrate and a second portion **1006** that connects with **1004** to cover the edge of a self-bonding FRP **402**. The first portion **1004** includes a backplate **1008** that connects to a substrate. A first portion of a back side of the backplate **1008** includes an adhesive **1010** that is used for securing the backplate **1008** of the first portion **1004** to the substrate. In one embodiment, the adhesive **1010** only covers a portion of the back side of the backplate **1008** (other areas may be covered). While covering a portion of the back side of the backplate **1008** with adhesive **1010** has

been described, it will be understood that lesser and greater areas on the back side of the backplate **1008** may be covered with the adhesive **1010**.

[0046] The front side of the backplate **1008** includes an edge marker **1012** marking the edge placement of a self-bonding FRP **402**. The front side of the backplate **1008** of the first portion **1004** has a female connector **1014** extending perpendicularly therefrom on one end of the backplate **1008**. The female connector **1014** includes a pair of latching members **1016** extending perpendicular from the backplate **1008**. Each of the pair of latching members **1016** define a ledge **1018** for securing a male connector **1020**. The latching members **1016** may flexibly move laterally to allow disengagement of the male connector **1020**.

[0047] The second portion **1006** of the trim accessory **1002** comprises a flexible plate **1022** and the male connector **1020** extending therefrom. A lateral plate **1021** extends from the flexible plate **1022** in the same direction as the male connector **1020**. The flexible plate **1022** has a slightly curved shape extending downward toward the backplate **1008** for holding an edge a self-bonding FRP **402** against the front surface of the backplate **1008**. The flexible nature of the flexible plate **1022** enables the corner trim accessory **1002** to accommodate multiple self-bonding FRP **402** thicknesses. The flexible nature further enables the flexible plate **1022** to provide a holding forces against the edge of the self-bonding FRPs **402**. The lateral plate **1021** extends from the flexible plate **1022** to provide an endcap for the edge of the self-bonding FRP **402**.

[0048] The male connector **1020** includes a horizontal member **1024** extending perpendicularly from the surface of the flexible plate **1022**. At the end of the horizontal member **1024**, opposite from the connection point with the flexible plate **1022**, an arrow shaped connector includes a first wing **1026** and a second wing **1028** enabling the male connector **1020** to latch into the female connector **1014**. When snapped into place, the first and second wings **1026**, **1028** click into place on the ledges **1018** of the female connector **1014** to secure the flexible plate **1022** to the backplate **1008**. This enables the endcap trim accessory **1002** to secure an edge of a self-bonding FRP **402** and provide an endcap protection for the edge. The endcap trim accessory **1002** speeds up install times and enhances ease of use of the self-bonding FRPs **402**.

[0049] FIG. **11** illustrates a corner trim accessory **1102** for speeding the installation process of self-bonding FRPs **402**. The corner trim accessory **1102** enables the interconnection of two adjacent self-bonding FRPs on an inside corner of a substrate. The corner trim accessory **1102** includes a first portion **1104** that connects to adjacent substrates forming an inside corner and a second portion **1106** that connects with and secures a pair of adjacent self-bonding FRPs that interconnect at the inside corner. The first portion **1104** includes a first backplate portion **1107** that connects to a first substrate forming an inside corner, a second backplate portion **1108** that is located along the inside corner and a third backplate portion **1109** that connects to a second substrate forming the inside corner. A first back side of the first backplate portion **1107** includes an adhesive **1110** that is used for securing the first backplate portion **1107** of the first portion **1104** to the first substrate. In one embodiment, the adhesive **1110** only covers a portion of the back side of the first backplate portion **1107** (other areas may be covered). The back side of the third backplate portion **1109** does not include any adhesive so the third backplate portion **1109** can be inserted under the self-bonding FRP **402** installed on the second substrate forming the inside corner. While covering a portion of the backside of the first backplate portion **1108** with adhesive **1110** has been described, it will be understood that lesser and greater areas on the back side of the first backplate portion **1107** and the third backplate portion **1109** and even the second backplate portion **1108** may be covered with the adhesive **1110**.

[0050] The front side of the first backplate portion **1107** and the third backplate portion **1109** includes an edge marker **1112** marking the edge placement of the first and second self-bonding FRPs **402** forming the inside corner. The front side of the second backplate portion **1108** of the first portion **1104** has a female connector **1114** extending perpendicularly therefrom. The female connector **1114** includes a pair of latching members **1116** extending perpendicular from the second

backplate portion **1108**. Each of the pair of latching members **1116** define a ledge **1118** for securing a male connector **1120**. The latching members **1116** may flexibly move laterally to allow disengagement of the male connector **1120**.

[0051] The second portion **1106** of the corner trim accessory **1102** comprises a flexible plate **1122** and the male connector **1120** extending therefrom. The flexible plate **1122** has a slightly curved shape with the edges curving away from the direction that the male connector extends for holding self-bonding FRPs **402** against the front surface of the first backplate portion **1107** and the second backplate portion **1109**. The flexible nature of the flexible plate **1122** enables the corner trim accessory **1102** to accommodate multiple self-bonding FRP **402** thicknesses. The flexible nature further enables the flexible plate **1122** to provide a holding forces against the edges of the self-bonding FRPs **402** intersecting at the inside corner.

[0052] The male connector **1120** includes a horizontal member **1124** extending perpendicularly from the outer curving surface of the flexible plate **1122**. At the end of the horizontal member **1124**, opposite from the connection point with the flexible plate **1122**, an arrow shaped connector includes a first wing **1126** and a second wing **1128** enabling the male connector **1120** to latch into the female connector **1114**. When snapped into place, the first and second wings **1126**, **1128** click into place on the ledges **1118** of the female connector **1114** to secure the flexible plate **1122** to the backplate. This enables the corner trim accessory **1102** to secure edges of self-bonding FRPs **402** forming the inside corner to their respective substrates. The corner trim accessory **1102** speeds up install times and enhances ease of use of the self-bonding FRPs **402** that are installed on inside corners.

[0053] Referring now to FIG. **12A**, there is illustrated a top view of a self-bonding FRP **402**. This view illustrates the manner in which the adhesive layer **406** and release liner layer **418** are laminated to the backside of the FRP **404**. Lamination of the adhesive layers **406** and release liner layers **408** to the backside of the FRP **404** can provide difficulties in aligning the edges of the FRP **404** with the edges of the adhesive layers **406** and tape layers **418**. In order to overcome this problem, the lamination process would have the dimensions of the adhesive layers **406** and **408** be slightly smaller than the dimensions of the FRP **404** along the edges **1200**. Thus, a boundary strip **1202** having no adhesive layer **406** or release liner layer **418** would be defined along the edges **1200** of the FRP **404**. The particular size of the boundary area with **1202** may be determined based upon design implementations. In further embodiments, the adhesive layers **406** and release liner layer **408** may cover the entire backside of the FRP **404** with no boundary strips. In a further embodiment, an RFID tag **1204** can be applied to a FRP **404** to detect temperature, humidity, moisture, strain, structural issues. The ultra-high frequency passive RFID tag **1204** would be applied to the FRP **404** before the application of the adhesive layer **406** and release liner layer **418**.

[0054] FIGS. **12B** and **12C** illustrate alternative embodiments wherein the adhesive layer **406** and the release liner layer **418** rather than covering substantially all of the backside of the FRP **404** may cover only portions of the backside of the FRP **404** in rectangular strips **1220** running in either the vertical direction (FIG. **12B**) or rectangular strips **1230** in the horizontal direction (FIG. **12C**). It will also be appreciated that adhesive layer **406** and release liner layer **418** portions may comprise any shape or be placed in any position on the backside of the FRP **404**. It will further be appreciated that the general shape of the self-adhesive FRP **402** need not be in a substantially rectangular or square configuration but may be in any shape according to the present invention.

[0055] FIG. **13** illustrates a flow diagram of the installation of the self-bonding FRP **402** to a substrate. Initially, at step **1302**, the endcap trim accessory **1002** is applied to an edge by placing the first portion **1004** on a substrate. Next, the self-bonding FRP **402** is placed at step **1304** upon the top surface of the first portion **1004** such that its edge of the FRP aligns with the edge marker **1012**. The self-bonding FRP **402** is then secured to the substrate by applying pressure or other adhesive activation means. The second portion **1006** of the endcap trim accessory **1002** is connected to the first portion **1004** in order to secure the edge of the self-bonding FRP **402** to

complete the endcap accessory installation at step **1306**.

[0056] Next, the division bar trim accessory **902** has one side of the first portion **904** inserted under the edge of the self-bonding FRP **402** at step **1308** that is currently mounted on the substrate such that the edge aligns with the edge marker **912**. The second self-bonding FRP **402** is then placed on the substrate with its edge placed at step **1310** on the top surface of the first portion **904** of the division bar trim accessory **902** such that the edge aligns with the edge marker **912**. The second self-bonding FRP **402** may then be secured to the substrate and the division bar trim accessory **902** by applying pressure or some other adhesive activation means. The second portion **906** of the division bar trim accessory **902** is then connected to the first portion **904** to secure each edge of the two adjacent self-bonding FRPs **402** to complete the installation of the trim assembly at step **1312**.

[0057] The further installation of self-bonding FRPs **402** may include placing additional adjacent self-bonding FRPs **402** that are interconnected via additional trim accessories **902** or protecting a stopping edge of the self-bonding FRP **402** using an additional endcap trim accessory **1002**. While any of these additional trim accessories **902** or endcap trim accessories **1002** may be utilized for purposes of the flow diagram of FIG. **13**, when an inside corner is reached this necessitates the use of a corner trim accessory **1102**. A portion of the first portion **1104** of the corner trim accessory **1102** is slid under an edge of the self-bonding FRP **402** currently mounted on a first substrate of the inside corner at step **1314** such that an edge of the self-bonding FRP **402** aligns with edge marker **1112**. The self-bonding FRP **402** forming the other substrate of the inside corner is then placed on the other top surface of the first portion **1104** of the corner trim accessory **1102** at step **1315**. The self-bonding FRP **402** is secured to the substrate and corner trim accessory **1102** by applying pressure or using some other type of adhesive activation means. The edge of the self-bonding FRP **402** is also aligned with the edge marker **1112**. The assembly of the corner trim accessory is completed at step **1316** by connecting the second portion **1106** with the first portion **1104** such that the flexible plate **1122** secures each edge of the self-bonding FRPs **402** forming the inside corner.

[0058] The described panels provide a number of advantages. They provide a moisture resistant panel that can be applied to walls or other structures. They provide repositioning ability until pressure is applied to the panel. This is because less surface area is connected to the substrate a panel is positioned on until the pressure is applied to the outer surface of the panel.

[0059] Although the preferred embodiment has been described in detail, it should be understood that various changes, substitutions and alterations can be made therein without departing from the spirit and scope of the invention as defined by the appended claims.

Claims

1. A self-bonding panel, comprising: a fiberglass reinforced panel having a front side and a backside; at least one adhesive layer applied to the backside of the fiberglass reinforced panel; and at least one release liner layer applied to a surface of the at least one adhesive layer opposite the fiberglass reinforced panel.
2. The self-bonding panel of claim 1, wherein the at least one adhesive layer further comprises: a first adhesive layer applied to the backside of the fiberglass reinforced panel; a carrier layer applied to the first adhesive layer opposite the fiberglass reinforced panel; and a second adhesive layer applied to the carrier layer opposite the first adhesive layer.
3. The self-bonding panel of claim 2, wherein the at least one adhesive layer further comprises: a performance layer on the second adhesive layer opposite the carrier layer; and a third adhesive layer located on the performance layer opposite the second adhesive layer.
4. The self-bonding panel of claim 2, wherein the carrier layer comprises a closed cell foam material.
5. The self-bonding panel of claim 1, wherein the at least one release liner layer comprises: a first release coating layer located on the at least one adhesive layer opposite the fiberglass reinforced

panel; a release liner located on the first release coating layer opposite the adhesive layer; and a second release coating layer located on the release liner opposite the first release coating layer.

6. The self-bonding panel of claim 1, wherein the at least one release liner layer comprises: a first release coating layer located on the at least one adhesive layer opposite the fiberglass reinforced panel; and a release liner located on the first release coating layer opposite the adhesive layer.

7. The self-bonding panel of claim 1, wherein the at least one release liner layer further comprises a texture layer, the texture layer configured to impart peaks and valleys into the at least one adhesive layer to mechanically enhance repositionability of the at least one adhesive layer.

8. The self-bonding panel of claim 1, wherein the at least one release liner layer is configured to be removed from the at least one adhesive layer to expose the at least one adhesive layer for connecting the self-bonding panel to a substrate.

9. The self-bonding panel of claim 1, wherein the at least one adhesive layer does not extend completely across a predetermined direction of the backside of the fiber reinforced panel such that a boundary strip is defined on opposing edges of the backside of the fiber reinforced panel that does not contain the adhesive layer.

10. The self-bonding panel of claim 1, wherein the at least one adhesive layer comprises a plurality of horizontal bands of adhesive layers extending across the backside of the fiber reinforced panel.

11. The self-bonding panel of claim 1, wherein the at least one adhesive layer comprises a plurality of vertical bands of adhesive layers extending across the backside of the fiber reinforced panel.

12. A system comprising: a plurality of self-bonding panels configured to connect to a substrate, each of the plurality of self-bonding panels further comprising: a fiberglass reinforced panel having a frontside and a backside; at least one adhesive layer applied to the backside of the fiberglass reinforced panel; at least one release liner layer applied to a surface of the at least one adhesive layer opposite the fiberglass reinforced panel; and at least one connection accessory for securing edges of the plurality of self-bonding panels to the surface.

13. The system of claim 12, wherein the at least one connection accessory comprises an endcap trim accessory, the endcap trim accessory comprising: a first member having a first side for connecting the endcap trim accessory to the substrate and a second side; a second flexible member configured to secure an edge of an associated self-bonding panel to the second side of the first member and to protect the edge of the associated self-bonding panel when connected to the first member; and a connector configured to securely connect the second flexible member to the first member.

14. The system of claim 12, wherein the at least one connection accessory comprises a division bar trim accessory, the division bar trim accessory comprising: a first member having a first side for connecting the division bar trim accessory to the substrate and a second side; a second flexible member configured to secure adjacent edges of two adjacent self-bonding panels to the second side of the first member when connected to the first member; and a connector configured to securely connect the second flexible member to the first member.

15. The system of claim 14, wherein the division bar trim accessory further includes an adhesive layer on at least a portion of the first side to connect division bar trim accessory to the substrate.

16. The system of claim 12, wherein the at least one connection accessory comprises an inside corner trim accessory, the inside corner trim accessory comprising: a first member having a first side for connecting the inside corner trim accessory to the substrate forming an inside corner and a second side; a second flexible member configured to secure adjacent edges of two adjacent self-bonding panels forming the inside corner when connected to the first member; and a connector configured to securely connect the second flexible member to the first member.

17. The system of claim 12, wherein the at least one adhesive layer each further comprise: a first adhesive layer applied to the backside of the fiberglass reinforced panel; a carrier layer applied to the first adhesive layer opposite the fiberglass reinforced panel; and a second adhesive layer applied to the carrier layer opposite the first adhesive layer.

- 18.** The system of claim 17, wherein the at least adhesive one layer further comprises: a performance layer on the second adhesive layer opposite the carrier layer; and a third adhesive layer located on the performance layer opposite the second adhesive layer.
- 19.** The system of claim 12, wherein the at least one adhesive layer does not extend completely across a predetermined direction of the backside of the fiber reinforced panel such that a boundary strip is defined on opposing edges of the backside of the fiber reinforced panel that does not contain the adhesive layer.
- 20.** The system of claim 12, wherein the at least one release liner layer comprises: a first release coating layer located on the at least one adhesive layer opposite the fiberglass reinforced panel; and a release liner located on the first release coating layer opposite the adhesive layer.
- 21.** The system of claim 12, wherein the at least one release liner layer further comprises a texture layer, the texture layer configured to impart peaks and valleys into the at least one adhesive layer to mechanically enhance repositionability of the at least one adhesive layer.
- 22.** The system of claim 12, wherein the at least one release liner layer is configured to be removed from the at least one adhesive layer to expose the at least one adhesive layer for connecting a self-bonding panel to the substrate.
- 23.** A self-bonding panel, comprising: a fiberglass reinforced panel having a front side and a backside; at least one adhesive layer applied to the backside of the fiberglass reinforced panel, wherein the at least one adhesive layer further comprises: a first adhesive layer applied to the backside of the fiberglass reinforced panel; a carrier layer applied to the first adhesive layer opposite the fiberglass reinforced panel; a second adhesive layer applied to the carrier layer opposite the first adhesive layer; at least one release liner layer applied to a surface of the at least one adhesive layer opposite the fiberglass reinforced panel, wherein the at least one release liner layer comprises: a first release coating layer located on the at least one adhesive layer opposite the fiberglass reinforced panel; a release liner located on the first release coating layer opposite the adhesive layer; and a second release coating layer located on the release liner opposite the first release coating layer; and a texture layer located within the at least one release liner layer configured to impart peaks and valleys into the at least one adhesive layer to mechanically enhance repositionability of the at least one adhesive layer.
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